



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -1 (Paper-1)

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DOS (Disk Operating System)

- MS-DOS means Microsoft Disk Operating System is a type of operating system for mostly used personal computers, developed by Microsoft.
- MS-DOS was the main operating system for IBM PC compatible personal computers during the 1980s and the early 1990s.
- MS-DOS is one of the oldest and widely used operating system.
- As this operating system is mainly deals with different disks like floppy, hard disk etc., it is called as Disk Operating System (DOS).
- When we start the computer, it makes the system to work in its internal parts like RAM, ROM and other peripherals. Here it will check whether the operating system has been loaded in it or not. If it is loaded, it will start further operations
- DOS is a set of computer programs, the major functions of which are file management, allocation of system resources, providing essential features to control hardware devices.
- DOS uses a command line, or text-based interface (CUI/CLI), that allows the user to type commands.
- This made the operating system difficult for novices(नौसिखिया) to use, which is why Microsoft later bundled the graphic-based(GUI) Windows operating system with DOS.
- DOS is a single-tasking, single-user operating system with a command-line interface.
- A and B refer to floppy disk drives, while C refers to the hard disk.

❖ Function of DOS

- It takes commands from the keyboard and interprets them.
- It shows all the files in the system.
- It creates new files and allots space for programme.
- It changes the name of a file in place of old name.
- It copies information in a floppy drive and hard disk.
- It helps in locating a file.
- It searches where the file is located in the disk.
- If we want the information in the file to be printed, it gives printout of the information.
- It hides the files and directories so as not to be seen by others.
- It permanently removes the file.

❖ Feature of DOS

- No Case sensitive (cls/CLS)
- It is a single user system.
- It controls program.
- It is machine independence.
- It manages (computer) files.
- It manages input and output system.
- It manages (computer) memory.
- It provides command processing facilities.
- It operates with Assembler.

❖ Difference b/w MS- DOS & Ms-Windows OS

DOS (Disk Operating System)	Windows OS
DOS is single user operating system	Windows is multitasking OS
It consume less power	It consume high power
It consume less memory	It consumes more memory
DOS does not support networking	Windows support networking
Complex to use	Easy and simple to use
DOS is command line OS	Windows is Graphical user Interface OS
Does not support multimedia like (games, movies, songs)	Support multimedia like (games, movies, songs)
Only one window opened at a time	Multiple windows can be opened at a time
DOS is free of cost	Windows original version is expensive

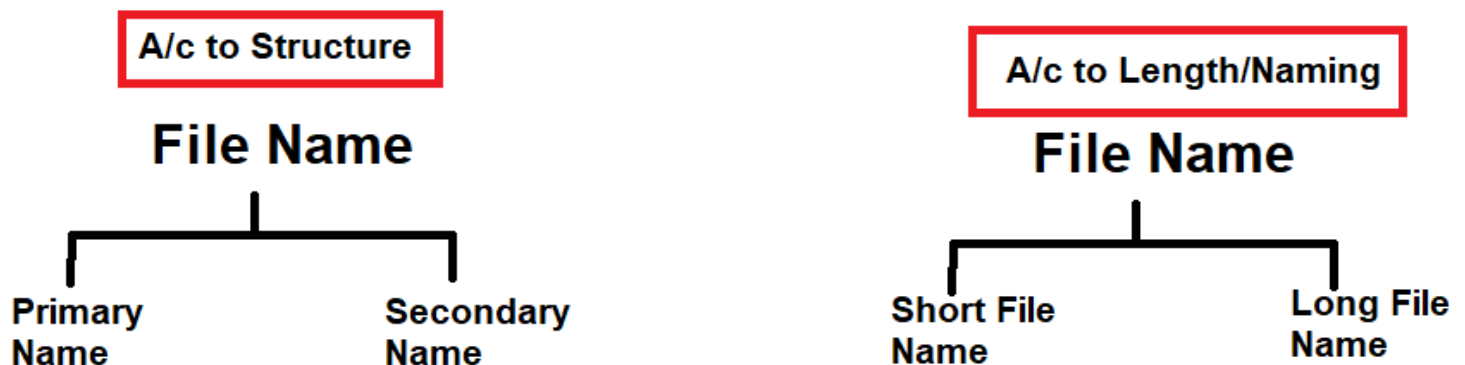
❖ What is a File

In computer terminology, file is a collection of text or data stored on a storage device, such as a Floppy Disk or Hard Disk.

❖ File Name

Every file have a filename which helps to recognize them.

Types of File Name



❖ Based to Structure

These are divided into two parts.

1. File name or Primary name
2. Extension or Secondary name.

1. Primary Name

- It is also called File name.
- Maximum 8 Characters store
- This is a actual/real name of file.

✓ **Example:-** BCA.pdf (here BCA is a file name/Primary name)

2. Secondary Name

- It is also called Extension.
- Maximum 3 characters store
- It is a type of file. (e.g. - .pdf, .jpg, .txt,.....)

✓ **Example:-** BCA.pdf (here .pdf is a Extension/Secondary name)

File Name = PrimaryName.SecondaryName

Example = BCA . pdf



❖ Based on Length/Naming = 2 types

1. Short File name (8.3 format)
2. Long File name

1. Short File Name

- It is also called DOS File Name. (Old DOS system support)
- Rules :-
 - (a) File Name : Max 8 Characters
 - (b) Extension : Max 3 Characters
 - (c) No space allowed
 - (d) Special Characters not allowed (/ \ : * ? " < > | ,.....)

✓ e.g. → bca.pdf ✓ bca-part1.pdf ✗ bihar sharif.jpg ✗
bihar.jpeg ✗

2. Long File Name

- It is support in Windows/Advanced DOS
- Rule:-
 - (a) Maximum 255 characters
 - (b) Spaces allowed
 - (c) Some Special characters allowed except slash, colon, star, question, quotes, Angle brackets, pipe → / \ : * ? " < > | (Reason → It characters is reserved for **System** Path,command,redirection & device handling)

✓ e.g. → ppu bca notes.docx ✓ bseb 12th result.pdf ✓
bca:part1.jpeg ✗

❖ Types of DOS Commands

DOS Commands are divided into 2 types:

1 Internal Commands

2 External Commands

1. Internal Command

- Those commands which are already stored in the “Command.Com” file or cmd.exe of DOS are known as internal commands.
- For example, CLS, VOL, TIME, DATE, COPY etc
- See all internal command type → C:\> help **OR** help | more

2. External Command

- Those commands which are not included in the command.com file of DOS rather included in other files of DOS are known as external commands. For example, TREE, FORMAT, MODE etc
- These are the .EXE or .COM programs located on our hard drive.

Difference b/w Internal & External Command

INTERNAL COMMAND

- These are those commands which are contained in files of MS-DOS.
- These are those functions that are built into the command interpreter.
- There is no need of any external file in computer to read internal MS-DOS command.
- Internal commands do not vary from system to system.
- These commands can be used as long as DOS is running on the system.

EXTERNAL COMMAND

- These are those commands which are not built in MS-DOS.
- These are those which are not included in the interpreter.
- There is a need of an external file in the computer to read external MS-DOS command.
- External command may vary from system to system.
- This means any two computers with same version of MS-DOS may have same internal commands, but may have different external commands.

➤ **Command Prompt** - A command prompt is the input field in a text-based user interface screen for an operating system (OS) or program.

➤ **Drive :-** Where all directory are store. OR A computer device that stores and retrieves information, data, files, programs, etc., from a disk. (e.g – A drive, B drive, C drive, D drive, E drive,...)

➤ **Directory [DIR] :-** Directory is a location for storing files on your computer. Directories are found in a hierarchical file system, such as Linux, MS-DOS, OS/2, and Unix.

➤ **Current Directory:-** Current directory is also called working directory or current working directory (CWD), the current directory is the directory or folder where you are currently working.

➤ **Root or Root directory:-** is the highest level in a directory hierarchy and includes all other directories under it. For example, in MS-DOS, the primary hard drive's root or root directory would be the C:\ directory.

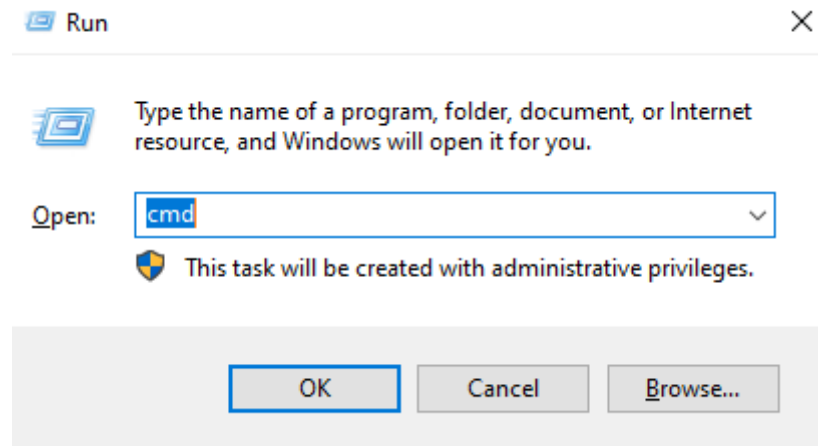
➤ **/ (Forward Slash) :-** It is called **Command Switch** OR **Command option** used in change function.

e.g – C:\> Dir /a , Dir /p, Dir/W, Dir/AH

- See all switch/option in related command then use help command
 - ✓ Syntax → C:\> help [command name]
 - ✓ Example :- C:\> help dir

➤ **How to Open Command Prompt**

1. **Step1 - Press window key and r (Window Key + R)**
2. **Step2 - Open Run dialog box Type cmd and click ok.**



➤ **Rules for naming files in Old DOS:-**

- A File name must have 1 to 8 characters.
- An optional extension of 1 to 3 characters – A period (.) between the name and extension name, if extension is used.
- All letters from A through Z (lower case letters are automatically transferred to uppercase), 0 to 9 numbers and special characters & symbols \$ # & @ ! ^ () _ - { } ' ~ are permitted in the file name.
- The control characters such as Esc, Del, or space bar cannot be used in the file name.
- The characters + = /(forward slash) [] :(colon) ; ? * < > : are not permitted.
- Each file name in a directory must be unique.

All keyboard symbols name in English

Symbol	Name
~	Tilde
!	Exclamation Mark
@	At Sign
#	Pound Sign or Hash
\$	Dollar Sign
%	Percent Sign
^	Caret
&	Ampersand
*	Asterisk
(	Left Parenthesis
)	Right Parenthesis
-	Hyphen or Dash
_	Underscore
=	Equal Sign
+	Plus Sign
{	Left Curly Brace or Left Brace
}	Right Curly Brace or Right Brace
,	,
[\]	Backslash
:	Colon
;	Semicolon
"	Double Quotation Mark or Quote
'	Single Quotation Mark or Apostrophe
<	Less Than Sign
>	Greater Than Sign
,	Comma
.	Period or Dot

Internal Commands :-

1. CLS Command

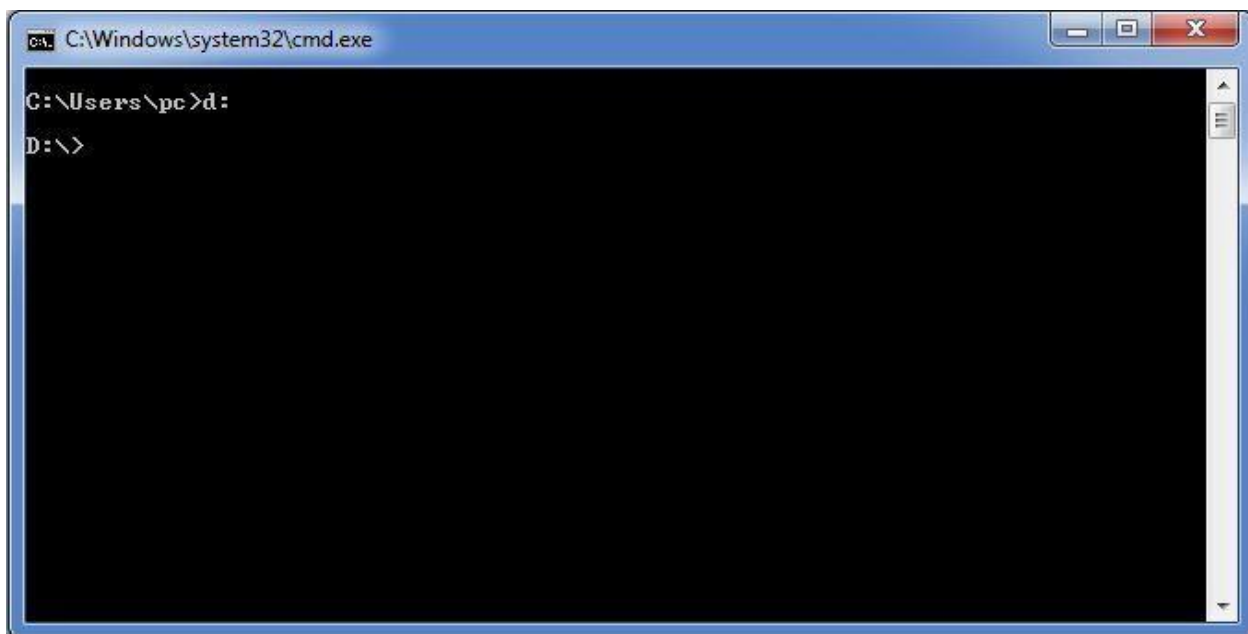
Work - To clear the screen.

On Screen/Syntax – C:\> cls press Enter key

2. Drive Change

Work – Change one drive to another drive

Syntax – c:\> d: press Enter key



3. DIR Command

Work - To view the directory and files.

Syntax - d:\> dir press Enter key

```
C:\Windows\system32\cmd.exe

D:\>dir
Volume in drive D is New Volume
Volume Serial Number is 740E-D5C5

Directory of D:\

01/12/2025  10:27 PM                102,033  49d34c118aa00bf99e7927c0f0aec1a9.jpg
12/17/2024  12:58 PM                145,218  ABHIS REG.pdf
12/11/2024  11:57 AM                593,623  Applications.pdf
12/17/2024  11:18 AM                379,066  AppPrint.aspx.pdf
09/24/2024  01:59 PM                <DIR>    B612
01/03/2025  04:40 PM                140,215  BPSC Portal.pdf
12/27/2024  12:44 PM                 29,604  Captur2.JPG
01/12/2025  09:30 PM                 64,316  Capture.JPG
01/12/2025  10:02 PM                 42,930  Capture1.JPG
01/12/2025  10:28 PM                 24,917  Capture12.JPG
01/02/2025  12:25 PM                354,373  Centralized Student Placement Management.
pdf
12/05/2024  01:35 PM                102,413  cnn.JPG
12/24/2024  05:44 PM                <DIR>    CUSTOMER
12/04/2024  10:45 AM                <DIR>    customer image
01/12/2025  10:30 PM                252,035  Disk Operating System.docx
10/29/2024  11:13 AM                <DIR>    GG
12/12/2024  11:27 AM                 29,491  HH.jpg
12/17/2024  12:44 PM                 80,643  HHJ.JPG
12/04/2024  08:40 PM                1,710,244  image-7.png
12/04/2024  08:42 PM                 10,812  images.jpg
12/11/2024  11:43 AM                2,407,076  IMPORTANT_FORMULAE_FOR_COMPETITIVE_EXAMS.
pdf
12/17/2024  12:54 PM                154,590  MINE REG.pdf
12/16/2024  05:26 PM                <DIR>    New folder
12/25/2024  01:01 PM                <DIR>    NRI
01/04/2025  08:51 AM                240,197  Patliputra University, Patna---.pdf
01/02/2025  05:02 PM                250,210  Patliputra University, Patna.pdf
12/18/2024  01:10 PM                126,136  payment_receipt.aspx 2.pdf
12/18/2024  01:14 PM                126,524  payment_receipt.aspx 3.pdf
12/18/2024  01:06 PM                129,474  payment_receipt.aspx.pdf
12/04/2024  08:45 PM                <DIR>    Presentation1
12/04/2024  08:47 PM                652,704  Presentation1.pptx
12/10/2024  12:21 PM                <DIR>    Presentation11
12/16/2024  05:13 PM                445,122  Print Form.pdf
12/04/2024  11:49 AM                <DIR>    RTPS
12/17/2024  12:59 PM                136,959  SAMBHAU REG.pdf
12/02/2024  04:08 PM                <DIR>    UIVEK
12/04/2024  08:10 PM                191,494  WhatsApp Image 2024-12-04 at 6.15.31 PM.j
peg
12/04/2024  08:11 PM                 74,156  WhatsApp Image 2024-12-04 at 6.15.32 PM <
1>.jpeg
12/04/2024  08:10 PM                 89,640  WhatsApp Image 2024-12-04 at 6.15.32 PM.j
peg
12/04/2024  08:19 PM                216,708  WhatsApp Image 2024-12-04 at 6.43.19 PM.j
peg
12/26/2024  01:17 PM                243,496  WhatsApp Image 2024-12-26 at 13.07.28.jpe
g
      31 File(s)          9,546,419 bytes
     10 Dir(s)    82,309,668,864 bytes free

D:\>
```

C:\> help dir

```

C:\>Help dir
Displays a list of files and subdirectories in a directory.

DIR [drive:][path][filename] [/A[:attributes]] [/B] [/C] [/D] [/L] [/N]
  [/O[:sortorder]] [/P] [/Q] [/R] [/S] [/T[:timefield]] [/W] [/X] [/4]

  [drive:][path][filename]
      Specifies drive, directory, and/or files to list.

  /A      Displays files with specified attributes.
  attributes  D Directories          R Read-only files
               H Hidden files        A Files ready for archiving
               S System files        I Not content indexed files
               L Reparse Points      O Offline files
               - Prefix meaning not

  /B      Uses bare format (no heading information or summary).
  /C      Display the thousand separator in file sizes. This is the
           default. Use /-C to disable display of separator.
  /D      Same as wide but files are list sorted by column.
  /L      Uses lowercase.
  /N      New long list format where filenames are on the far right.
  /O      List by files in sorted order.
  sortorder  N By name (alphabetic)    S By size (smallest first)
               E By extension (alphabetic) D By date/time (oldest first)
               G Group directories first - Prefix to reverse order

  /P      Pauses after each screenful of information.
  /Q      Display the owner of the file.
  /R      Display alternate data streams of the file.
  /S      Displays files in specified directory and all subdirectories.
  /T      Controls which time field displayed or used for sorting
  timefield  C Creation
               A Last Access
               W Last Written

  /W      Uses wide list format.
  /X      This displays the short names generated for non-8dot3 file
           names. The format is that of /N with the short name inserted
           before the long name. If no short name is present, blanks are
           displayed in its place.
  /4      Displays four-digit years

Switches may be preset in the DIRCMD environment variable. Override
preset switches by prefixing any switch with - (hyphen)--for example, /-W.

```

4. Date Command

Work - To View and change the date.While only see date use date/t

Syntax – view & change – c:\ > date press Enter key

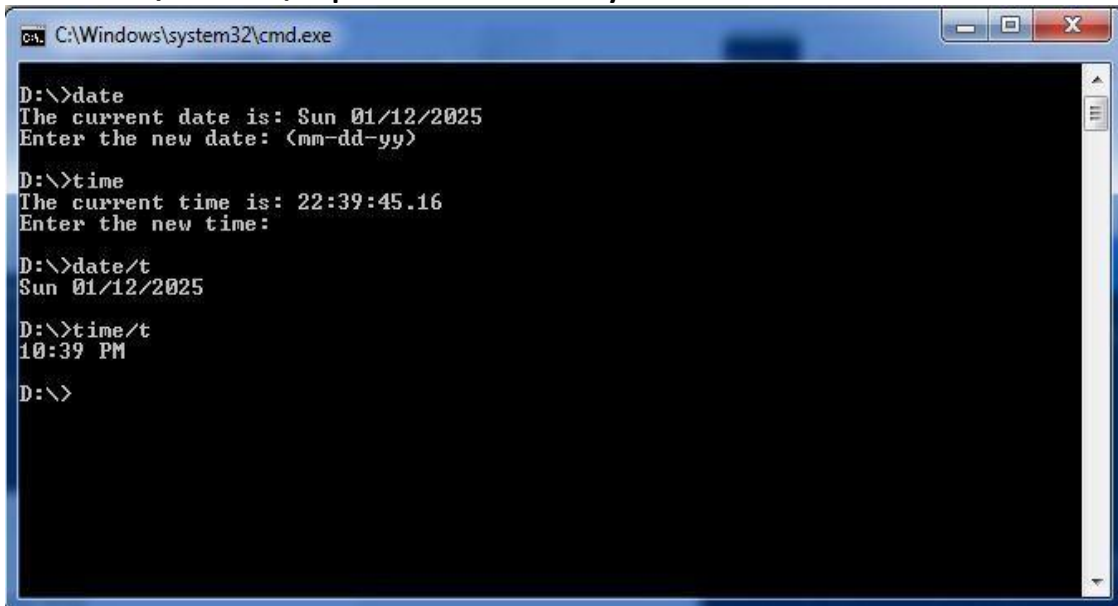
Only see – c:\> date/t press Enter key

5. TIME Command

Work - To view and change the time.While only see time use time/t

Syntax – view & change – c:\ > time press Enter key

Only see – c:\> time/t press Enter key



```
C:\Windows\system32\cmd.exe

D:\>date
The current date is: Sun 01/12/2025
Enter the new date: <mm-dd-yy>

D:\>time
The current time is: 22:39:45.16
Enter the new time:

D:\>date/t
Sun 01/12/2025

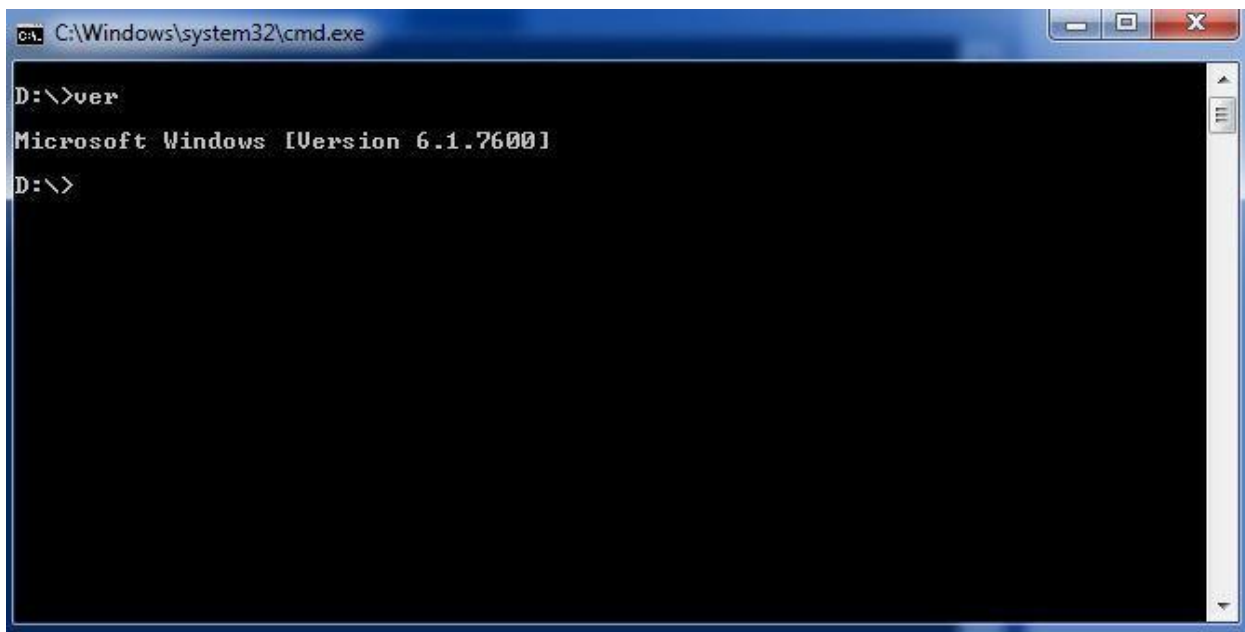
D:\>time/t
10:39 PM

D:\>
```

6. VER Command (version)

Work - Version numbers indicates that which edition of DOS we are working on

Syntax – c:\>ver press enter key



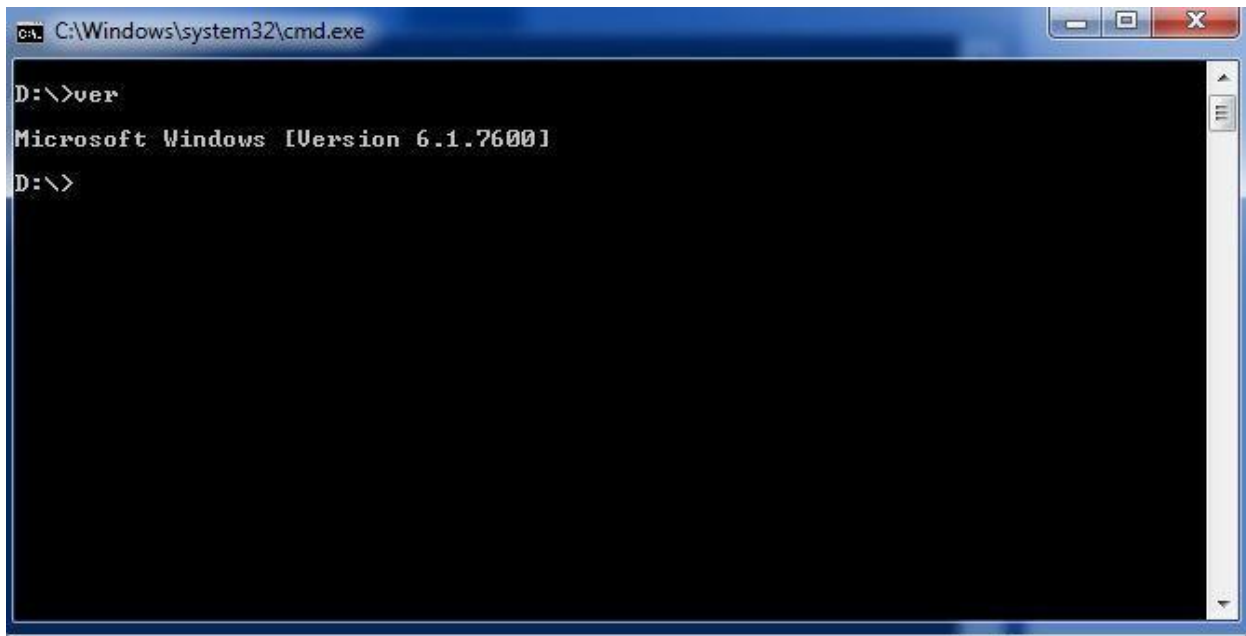
```
C:\Windows\system32\cmd.exe

D:\>ver
Microsoft Windows [Version 6.1.7600]

D:\>
```

7. VOL Command (Volume)

Work - Displays the disk volume label and serial number, if it exist.



```
C:\Windows\system32\cmd.exe
D:\>ver
Microsoft Windows [Version 6.1.7600]
D:\>
```

8. Color Command

Work - The command takes in input 2 hexadecimal digits (2nd digit optional) where the first digit corresponds to the background color and the second digit corresponds to the foreground/text color.

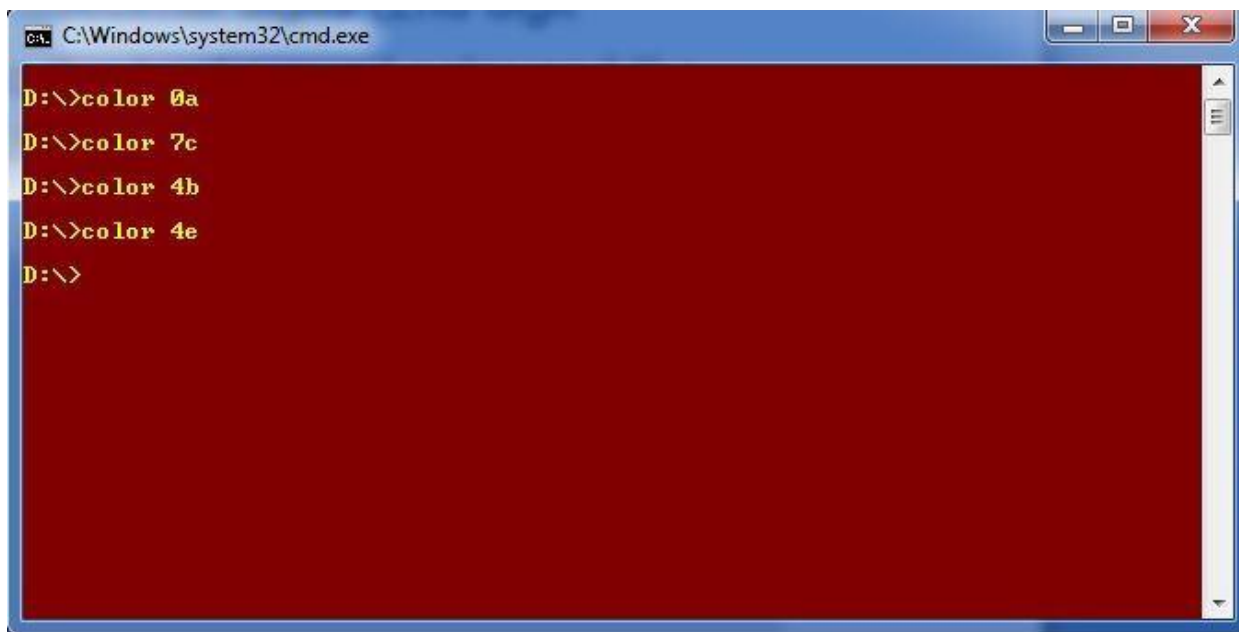
Color code menu view command – c:/> color halp OR color /?

Syntax:

Color (Background)(Foreground)

0 = Black	8 = Gray
1 = Blue	9 = Light Blue
2 = Green	A = Light Green
3 = Aqua	B = Light Aqua
4 = Red	C = Light Red
5 = Purple	D = Light Purple
6 = Yellow	E = Light Yellow
7 = White	F = Bright White

Example :- color 4e



```
C:\Windows\system32\cmd.exe

D:\>color 0a
D:\>color 7c
D:\>color 4b
D:\>color 4e
D:\>
```